# Bookings & Bowl Games

## Quantifying Hotel Impact of College Football Post-season

By Benjamin Taraba and Partick Obal

# What is the economic impact of hosting a college football bowl game?

# The Data

- We have daily hotel data from Jan 2020 to Jan 2026
- In 5 different locations
  - Atlanta, GA
  - New Orleans, LA
  - Miami, FL
  - Arlington, TX
  - Pasadena, CA
  - Phoenix, AZ

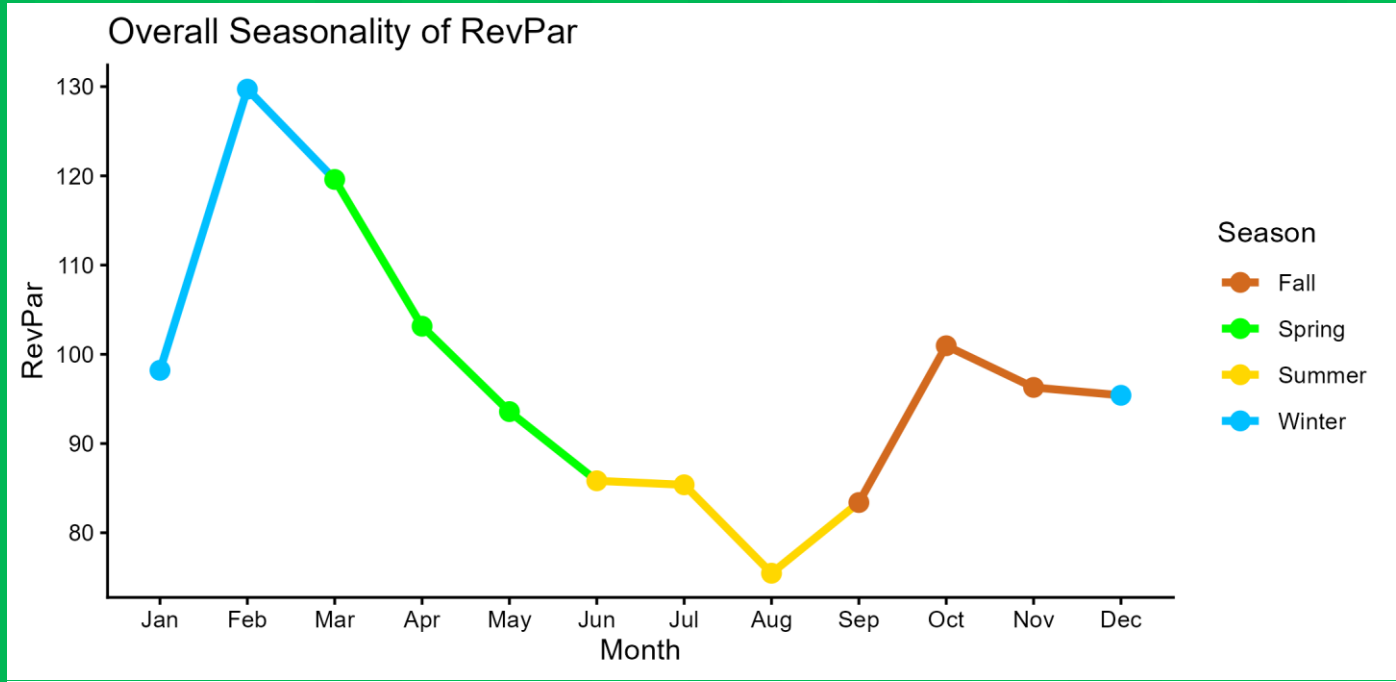
- We have daily Hotel Metrics like: Demand, Supply, Revenue, RevPar, etc.
- 38 different Bowl Games

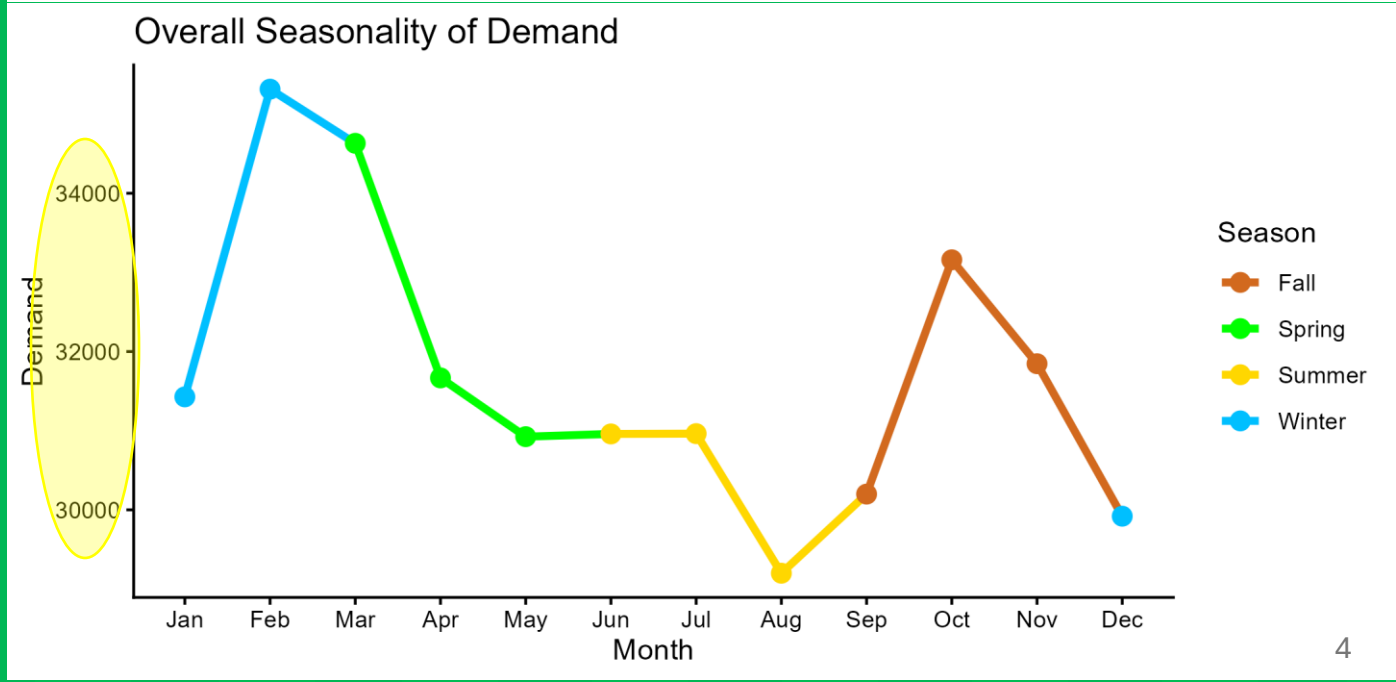
# Seasonality

- $RevPar = \frac{Total\ Room\ Revenue}{n\ of available\ rooms}$

  - Clear seasonality with a peak in winter and a trough in summer

  - Warm weather locations



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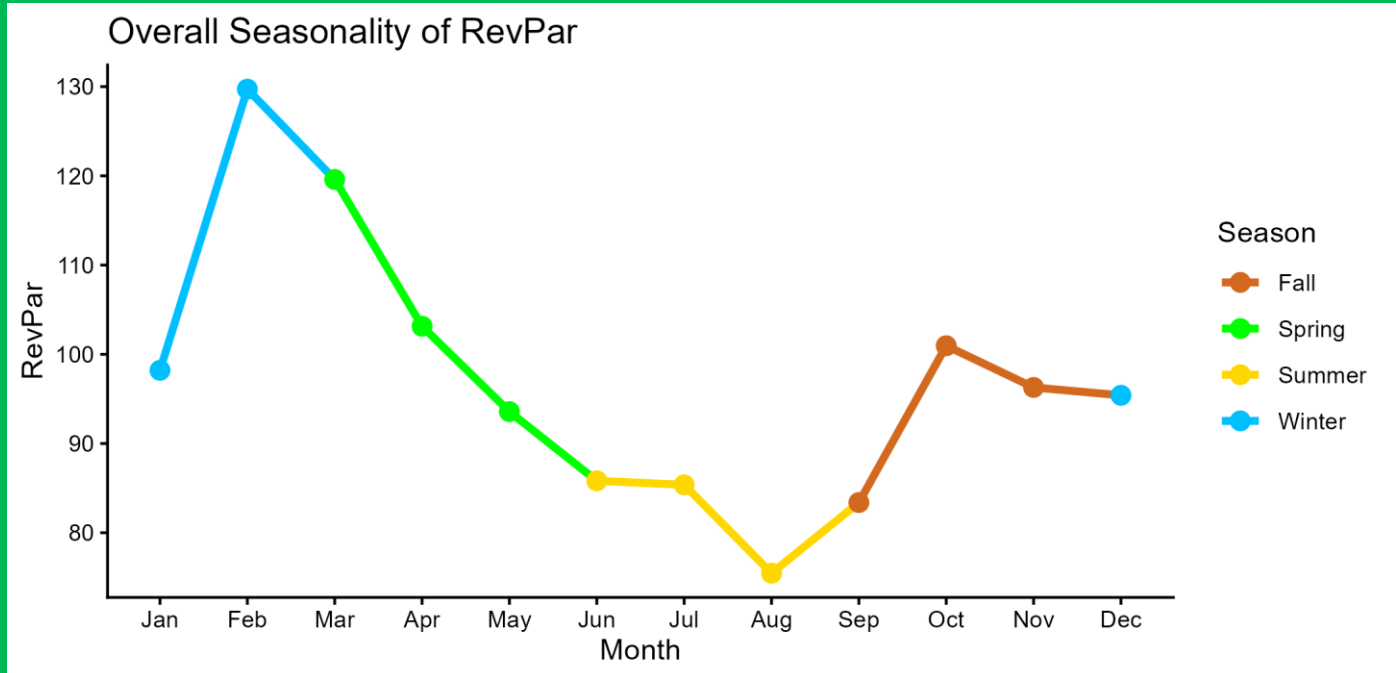
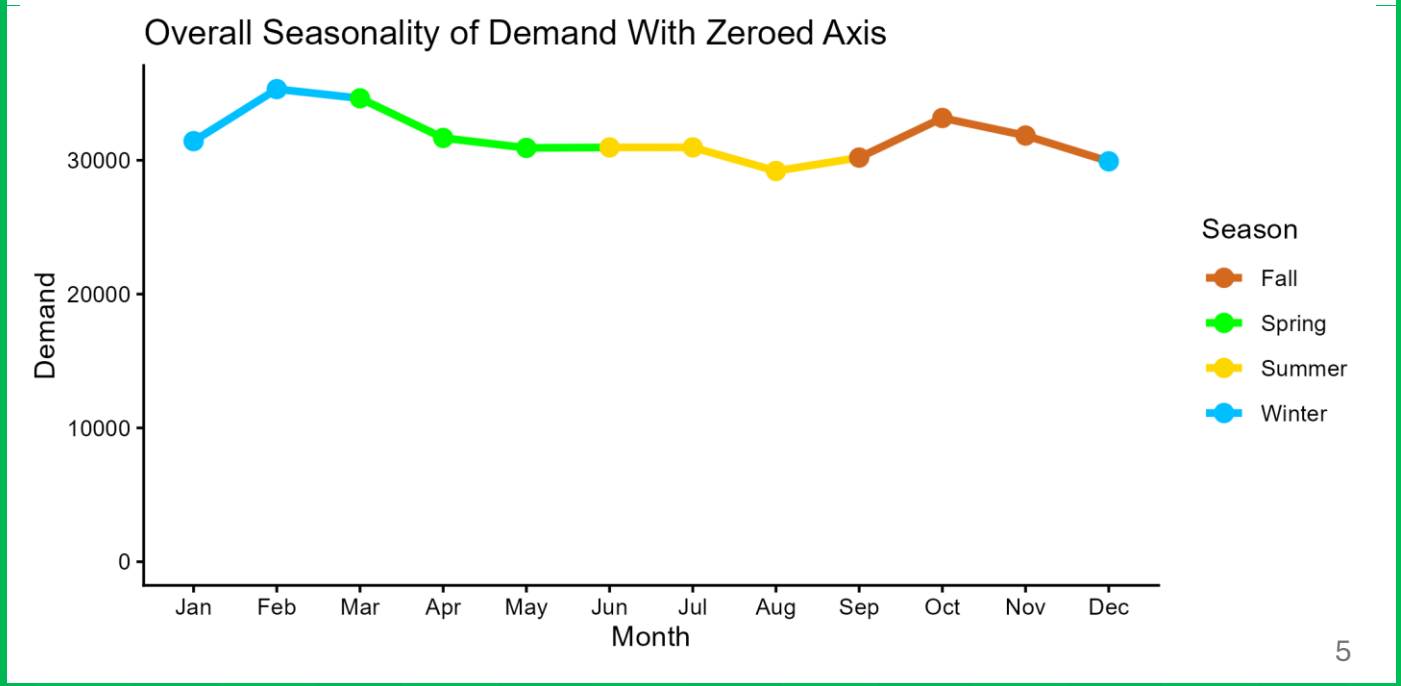
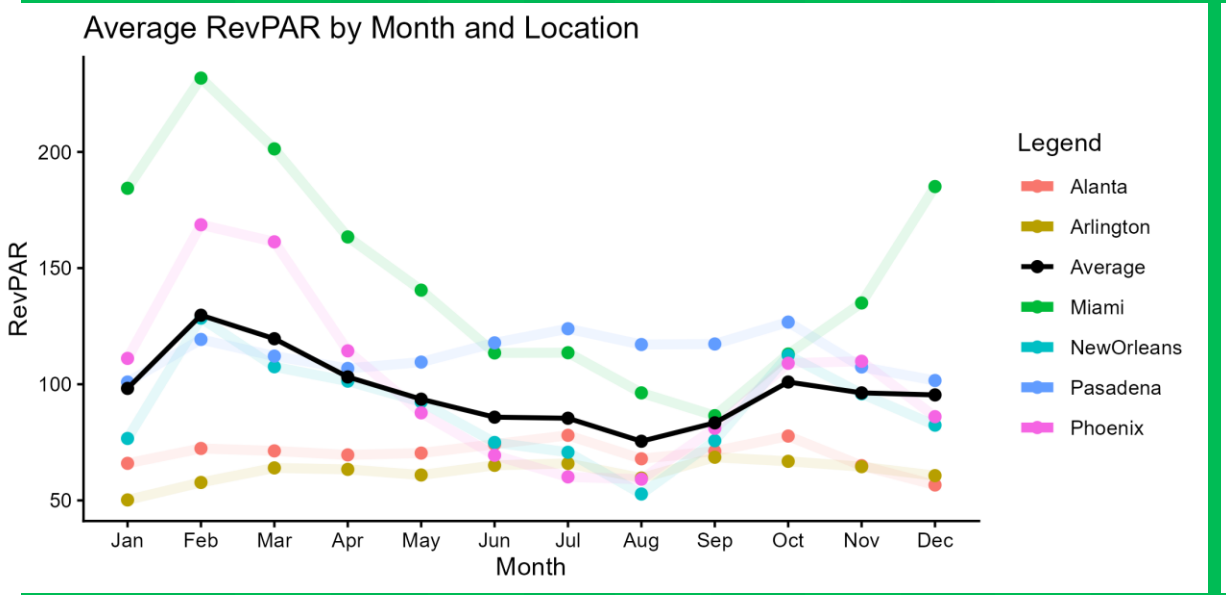
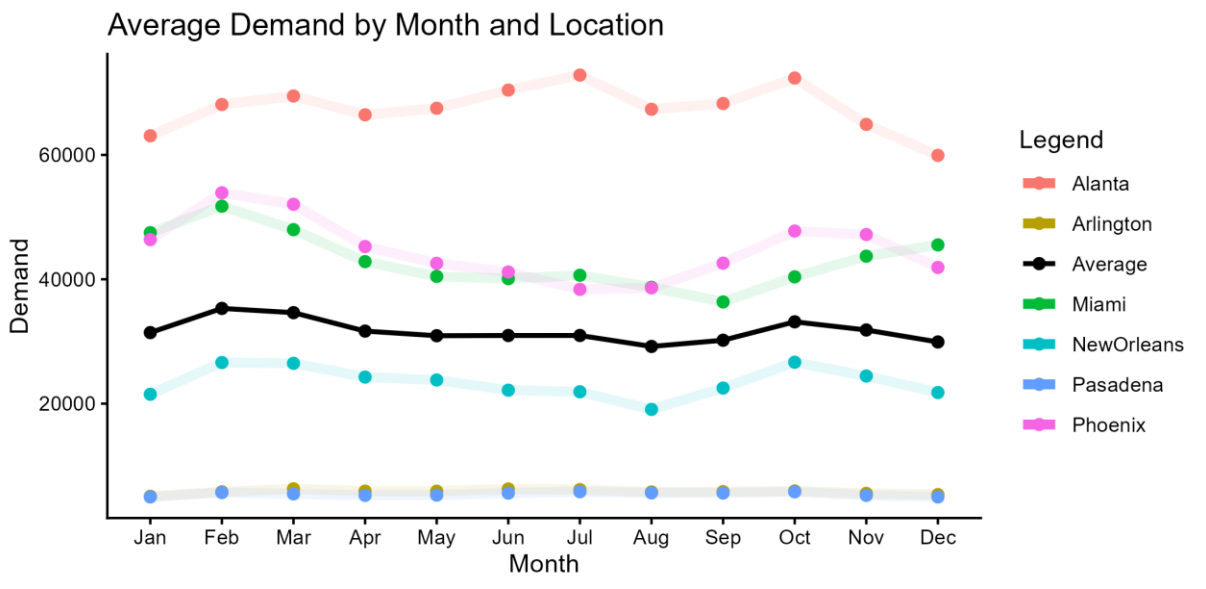


# Is there variety within cities?

- Before we do any significance testing, we want to see if there is a large variance between cities
- Lots of city variance in Demand and in RevPar.
- Some seasonality variance too



# Baseline RevPar

- RevPar baselines statistically differ
- Tukey's HSD proves that all the baselines are statistically different from each other

```
               Df   Sum Sq Mean Sq F value Pr(>F)
Location        5 11081536 2216307   873.2 <2e-16 ***
Residuals   13500 34264303    2538
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

| Mean RevPar by City | |
|---|---|
| Atlanta | 70 |
| Arlington | 62 |
| Miami | 148 |
| New Orleans | 89 |
| Pasadena | 113 |
| Phoenix | 102 |

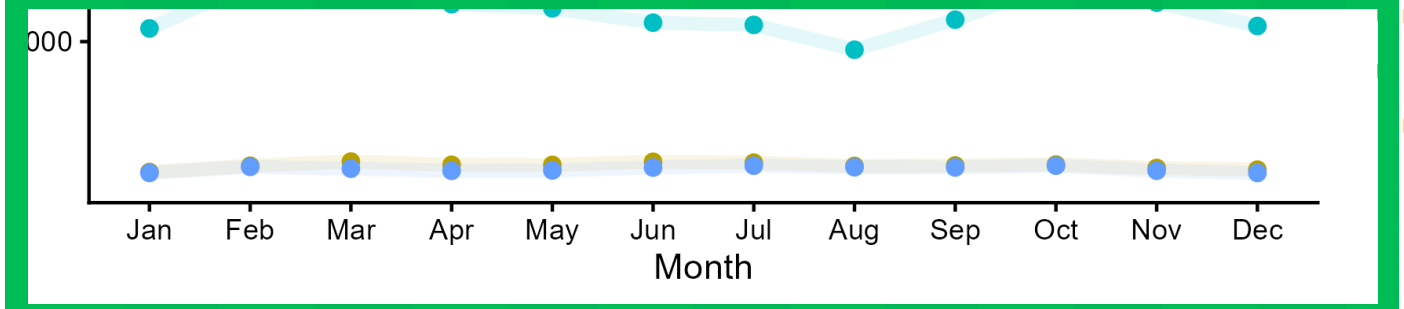
```
Fit: aov(formula = RevPar ~ Location, data = dat)

$Location
                          diff        lwr       upr    p adj
Atlanta-Arlington      7.909005   3.629005  12.18901 2.1e-06
Miami-Arlington       86.167170  81.887170  90.44717 0.0e+00
NewOrleans-Arlington  27.272251  22.992251  31.55225 0.0e+00
Pasadena-Arlington    51.195652  46.915652  55.47565 0.0e+00
Phoenix-Arlington     39.999942  35.719942  44.27994 0.0e+00
Miami-Atlanta         78.258165  73.978164  82.53817 0.0e+00
NewOrleans-Atlanta    19.363246  15.083245  23.64325 0.0e+00
Pasadena-Atlanta      43.286647  39.006646  47.56665 0.0e+00
Phoenix-Atlanta       32.090937  27.810936  36.37094 0.0e+00
NewOrleans-Miami     -58.894919 -63.174919 -54.61492 0.0e+00
Pasadena-Miami       -34.971518 -39.251518 -30.69152 0.0e+00
Phoenix-Miami        -46.167228 -50.447228 -41.88723 0.0e+00
Pasadena-NewOrleans   23.923401  19.643401  28.20340 0.0e+00
Phoenix-NewOrleans    12.727691   8.447691  17.00769 0.0e+00
Phoenix-Pasadena     -11.195710 -15.475710  -6.91571 0.0e+00
```

# Baseline Demand

- Demand baselines statistically differ
- Tukey's HSD proves that all the baselines are statistically different from each other except Pasadena and Arlington

```
               Df    Sum Sq   Mean Sq F value Pr(>F)
Location        5 6.782e+12 1.356e+12   15501 <2e-16 ***
Residuals   13500 1.181e+12 8.751e+07
```

| Mean Demand by City | |
|---|---|
| Atlanta | 6749 |
| Arlington | 5828 |
| Miami | 43130 |
| New Orleans | 23427 |
| Pasadena | 5454 |
| Phoenix | 44896 |




```
Fit: aov(formula = Demand ~ Location, data = dat)

$Location
                          diff         lwr         upr     p adj
Atlanta-Arlington     61669.5580  60874.8433  62464.2727 0.0000000
Miami-Arlington       37302.3101  36507.5954  38097.0248 0.0000000
NewOrleans-Arlington  17599.0800  16804.3652  18393.7947 0.0000000
Pasadena-Arlington     -374.1661  -1168.8809    420.5486 0.7615552
Phoenix-Arlington     39068.1035  38273.3888  39862.8182 0.0000000
Miami-Atlanta        -24367.2479 -25161.9626 -23572.5332 0.0000000
NewOrleans-Atlanta   -44070.4780 -44865.1927 -43275.7633 0.0000000
Pasadena-Atlanta     -62043.7241 -62838.4388 -61249.0094 0.0000000
Phoenix-Atlanta      -22601.4545 -23396.1692 -21806.7397 0.0000000
NewOrleans-Miami     -19703.2301 -20497.9448 -18908.5154 0.0000000
Pasadena-Miami       -37676.4762 -38471.1909 -36881.7615 0.0000000
Phoenix-Miami          1765.7934    971.0787   2560.5081 0.0000000
Pasadena-NewOrleans  -17973.2461 -18767.9608 -17178.5314 0.0000000
Phoenix-NewOrleans    21469.0235  20674.3088  22263.7383 0.0000000
Phoenix-Pasadena      39442.2697  38647.5549  40236.9844 0.0000000
```

# Going further

- Linear Effect Models (LM) – Estimates only fixed effects by assuming all variables are independent and differences are fixed
  - RevPar and Demand statistically differ by cities
- Linear Mixed Effect Models (LME) – Estimates both fixed and random effects, allowing for city baseline variation
  - Accounts for correlated observations

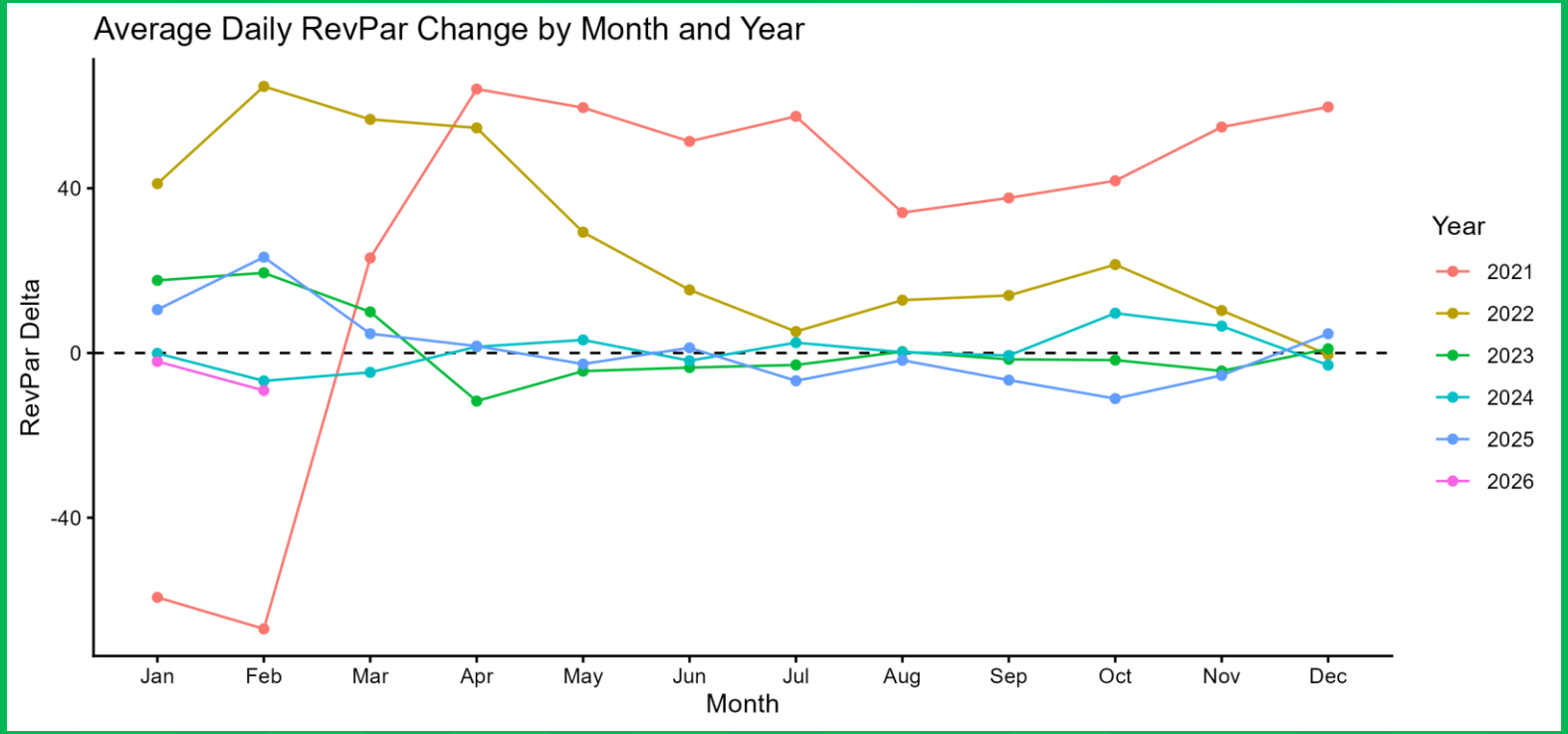
# Seasonality Throughout Years

- Years differing are expected as growth is expected
- However, years changing at drastically differing rates can affect overall month averages and make them less realizable.
- LME shows that the years statistically differ

```
Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2733 -0.6343 -0.0844  0.5008 15.9107

Random effects:
 Groups   Name        Variance Std.Dev.
 Location (Intercept)  983.9   31.37
 Residual             1612.2   40.15
Number of obs: 13506, groups:  Location, 6

Fixed effects:
                            Estimate Std. Error         df t value Pr(>|t|)
(Intercept)                  69.3780    12.9092     5.1566   5.374 0.002735 **
factor(Month)2               31.6924     1.6110 13477.0000  19.672  < 2e-16 ***
factor(Month)3               24.8284     1.6707 13477.0000  14.861  < 2e-16 ***
factor(Month)4                7.9940     1.6850 13477.0000   4.744 2.12e-06 ***
factor(Month)5               -1.4994     1.6707 13477.0000  -0.897 0.369482
factor(Month)6               -8.9232     1.6850 13477.0000  -5.296 1.21e-07 ***
factor(Month)7               -9.8150     1.6707 13477.0000  -5.875 4.33e-09 ***
factor(Month)8              -19.4689     1.6707 13477.0000 -11.653  < 2e-16 ***
factor(Month)9              -11.5439     1.6850 13477.0000  -6.851 7.66e-12 ***
factor(Month)10               5.8213     1.6707 13477.0000   3.484 0.000495 ***
factor(Month)11               1.3836     1.6850 13477.0000   0.821 0.411616
factor(Month)12               0.3971     1.6707 13477.0000   0.238 0.812116
factor(DayOfWeek)Monday     -32.3096     1.2929 13477.0000 -24.989  < 2e-16 ***
factor(DayOfWeek)Saturday     7.2070     1.2919 13477.0000   5.579 2.47e-08 ***
factor(DayOfWeek)Sunday     -33.2531     1.2929 13477.0000 -25.719  < 2e-16 ***
factor(DayOfWeek)Thursday   -20.9527     1.2919 13477.0000 -16.219  < 2e-16 ***
factor(DayOfWeek)Tuesday    -26.3974     1.2929 13477.0000 -20.417  < 2e-16 ***
factor(DayOfWeek)Wednesday  -23.7839     1.2919 13477.0000 -18.410  < 2e-16 ***
factor(Year)2021             30.1433     1.2126 13477.0000  24.859  < 2e-16 ***
factor(Year)2022             56.9604     1.2126 13477.0000  46.975  < 2e-16 ***
factor(Year)2023             58.5213     1.2126 13477.0000  48.262  < 2e-16 ***
factor(Year)2024             59.1271     1.2117 13477.0000  48.796  < 2e-16 ***
factor(Year)2025             59.9275     1.2126 13477.0000  49.422  < 2e-16 ***
factor(Year)2026             65.7461     2.4328 13477.0000  27.025  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

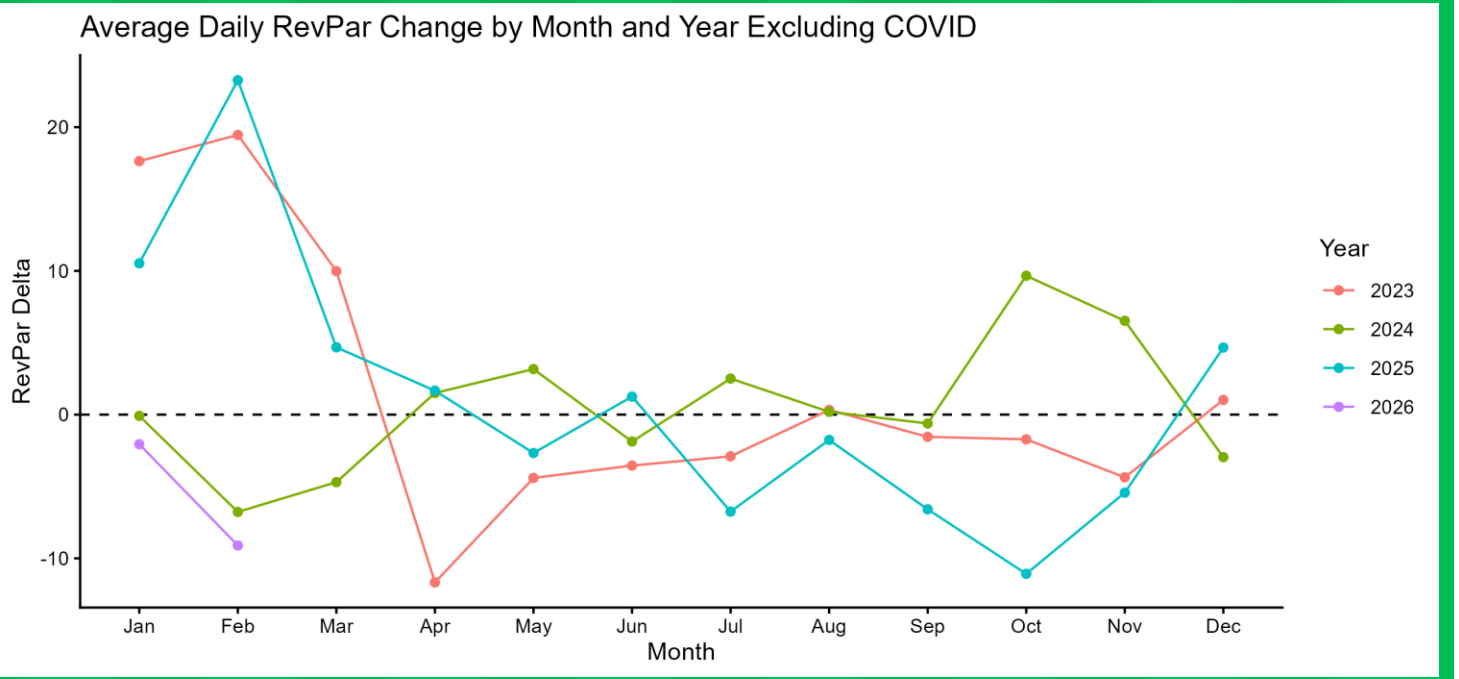
# Do they change at different rates?

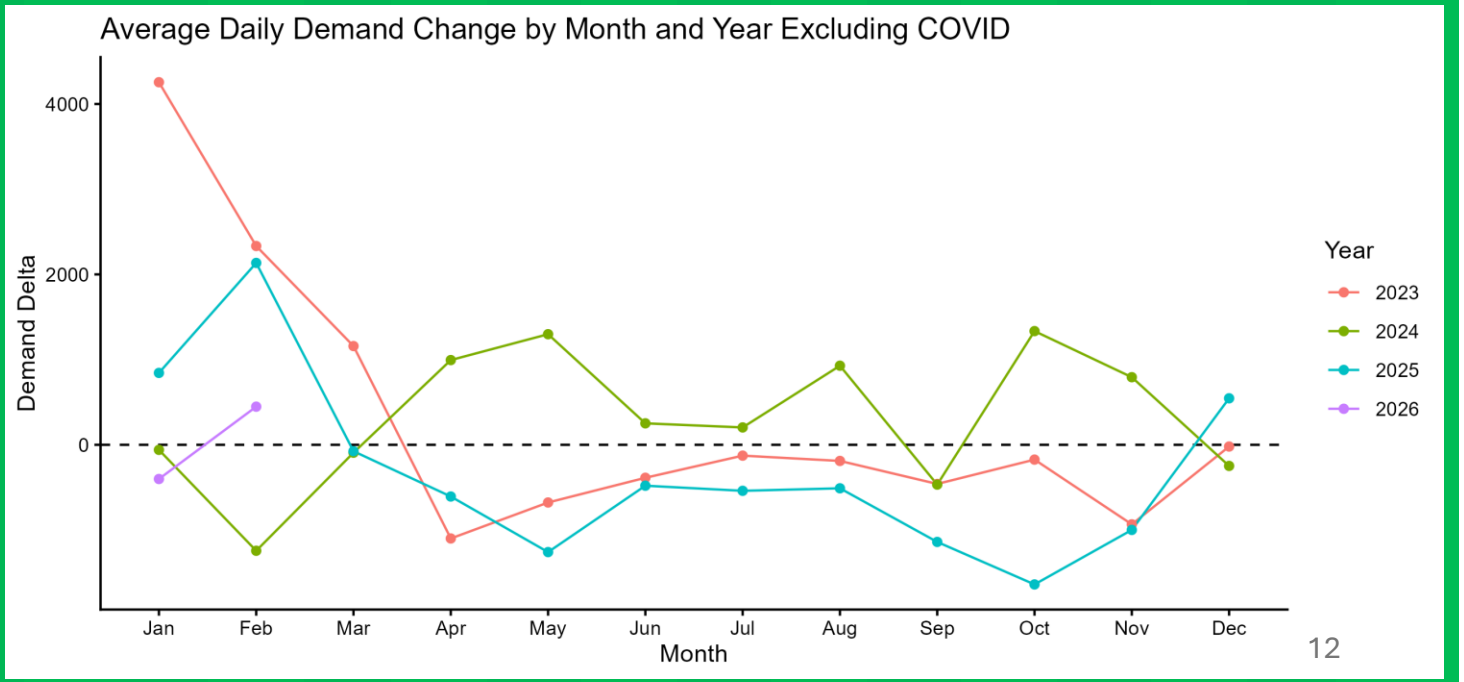
- Yes, large COVID impacts
- January Example
- 2021: -59.97
- 2022: -17.82
- 2023: +0.90
- 2024: +0.52
- 2025: +9.84
- 2026: +6.78



# Excluding Covid

- Monthly change in daily RevPAR and Demand varies not only by magnitude but also by direction
- Relying solely on monthly rates can mask year-specific trends and weaken prediction.



# Days of the Week

- We also see that the day of the week has seasonality, with Saturday being the best, followed by Friday, and Sunday being the worst

- Looking at DoW splits rather than weekday/end splits shows more of the real story.

```
Scaled residuals:
    Min      1Q  Median      3Q     Max
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Random effects:
 Groups   Name        Variance Std.Dev.
 Location (Intercept)  983.9   31.37
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factor(Year)2025             59.9275     1.2126 13477.0000  49.422  < 2e-16 ***
factor(Year)2026             65.7461     2.4328 13477.0000  27.025  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

# What does this all Mean?

- Clear Seasonality Winter Peaks and summer troughs
- Cities differ significantly within RevPar and Demand baselines
- Year Effects Matter, COVID and months grow independently
- Dow: Friday & Saturday are the strongest, while Sunday is the weakest.
- LME captures all this seasonality and variation over LM

# Event Effects

# Events Effects on RevPar

- Accounting for Seasonality, Year differences, and cities, our Linear Mixed Effects model found
- Average baseline of **$68.02** RevPar
- Average effect of an event of **$70.54** RevPar.
- With 95% confidence, a lower bound of $57.71 and a higher bound of $83.37

# Event effects on Rev Par by City

- Rev Par by City mainly has increases, like in
  - Miami 153.08
  - Pasadena 142.91
  - New Orleans 77.58
  - Arlington 65.76

Though I don’t have much effect on Rev par in

  - Atlanta -8.85
  - Phoenix -7.47

- Cities could experience negative or RevPar when events are in town because:
  - They might have to cut prices to stay competitive
  - Doesn't add tourist, replaces regular traffic
  - Time of year could already be popular like Phoenix and Atlanta in the Winter

# Demand

- After controlling for seasonality, year, day of week, and location, event days did not show a statistically significant effect on demand but estimated a positive effect of 2046 more units demanded.

```
REML criterion at convergence: 279154.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.9842 -0.5316 -0.0327  0.5760  4.1300

Random effects:
 Groups   Name        Variance  Std.Dev.
 Location (Intercept) 602572807 24547
 Residual              56668634  7528
Number of obs: 13506, groups:  Location, 6

Fixed effects:
                          Estimate Std. Error        df t value Pr(>|t|)
(Intercept)               24247.155  10026.125     5.008   2.418 0.060154 .
Event                      2045.910   1232.418 13476.000   1.660 0.096922 .
```

# Supply

- We tested Supply, and neither LME nor LM found an event effect on supply.
- The only significant effects were Year, City, and some Months.
- With a 99.45% Adjusted R^2

```
Call:
lm(formula = Supply ~ Event * Location + factor(Month) + factor(Year) +
    factor(DayOfWeek), data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-26846.7   -987.1    -66.4   1215.8   5614.8

Coefficients:
                          Estimate Std. Error  t value Pr(>|t|)
(Intercept)                5538.330    120.366   46.013  < 2e-16 ***
Event                      -550.494   1011.231   -0.544 0.586189
LocationAtlanta           100933.229     79.582 1268.293  < 2e-16 ***
LocationMiami              52747.942     79.582  662.813  < 2e-16 ***
LocationNewOrleans         32038.455     79.591  402.539  < 2e-16 ***
LocationPasadena           -1536.756     79.582  -19.310  < 2e-16 ***
LocationPhoenix            59878.105     79.582  752.407  < 2e-16 ***
factor(Month)2              -446.237    107.328   -4.158 3.23e-05 ***
factor(Month)3               174.991    111.257    1.573 0.115774
factor(Month)4             -1580.980    112.208  -14.090  < 2e-16 ***
factor(Month)5             -1548.302    111.257  -13.916  < 2e-16 ***
factor(Month)6              -767.631    112.207   -6.841 8.19e-12 ***
factor(Month)7              -291.632    111.258   -2.621 0.008771 **
factor(Month)8               -58.335    111.257   -0.524 0.600064
factor(Month)9                15.864    112.207    0.141 0.887568
factor(Month)10              190.397    111.257    1.711 0.087044 .
factor(Month)11              256.980    112.207    2.290 0.022024 *
factor(Month)12              424.420    111.120    3.819 0.000134 ***
factor(Year)2021            3681.208     80.528   45.713  < 2e-16 ***
factor(Year)2022            4725.007     80.524   58.679  < 2e-16 ***
factor(Year)2023            5407.077     80.517   67.154  < 2e-16 ***
factor(Year)2024            5797.798     80.464   72.055  < 2e-16 ***
factor(Year)2025            6358.545     80.516   78.973  < 2e-16 ***
factor(Year)2026            6517.722    161.589   40.335  < 2e-16 ***
factor(DayOfWeek)Monday       15.111     85.886    0.176 0.860342
factor(DayOfWeek)Saturday     -1.162     85.825   -0.014 0.989197
factor(DayOfWeek)Sunday       -1.538     85.889   -0.018 0.985713
factor(DayOfWeek)Thursday      1.477     85.811    0.017 0.986268
factor(DayOfWeek)Tuesday      17.373     85.892    0.202 0.839711
factor(DayOfWeek)Wednesday     1.846     85.803    0.022 0.982838
Event:LocationAtlanta        693.338   1485.936    0.467 0.640794
Event:LocationMiami         1167.945   1486.123    0.786 0.431940
Event:LocationNewOrleans     680.988   1428.808    0.477 0.633646
Event:LocationPasadena       -33.977   1487.429   -0.023 0.981776
Event:LocationPhoenix         96.916   1487.068    0.065 0.948038
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2666 on 13471 degrees of freedom
Multiple R-squared:  0.9945,    Adjusted R-squared:  0.9945
F-statistic: 7.16e+04 on 34 and 13471 DF,  p-value: < 2.2e-16
```

# Occupancy Rate

- Average baseline of 50.8% occupancy rate
- LME found a statistically significant event effect of 8.24 percentage points
  - p value: 7.10e-06
- 95% CL between 4.64 - 11.83

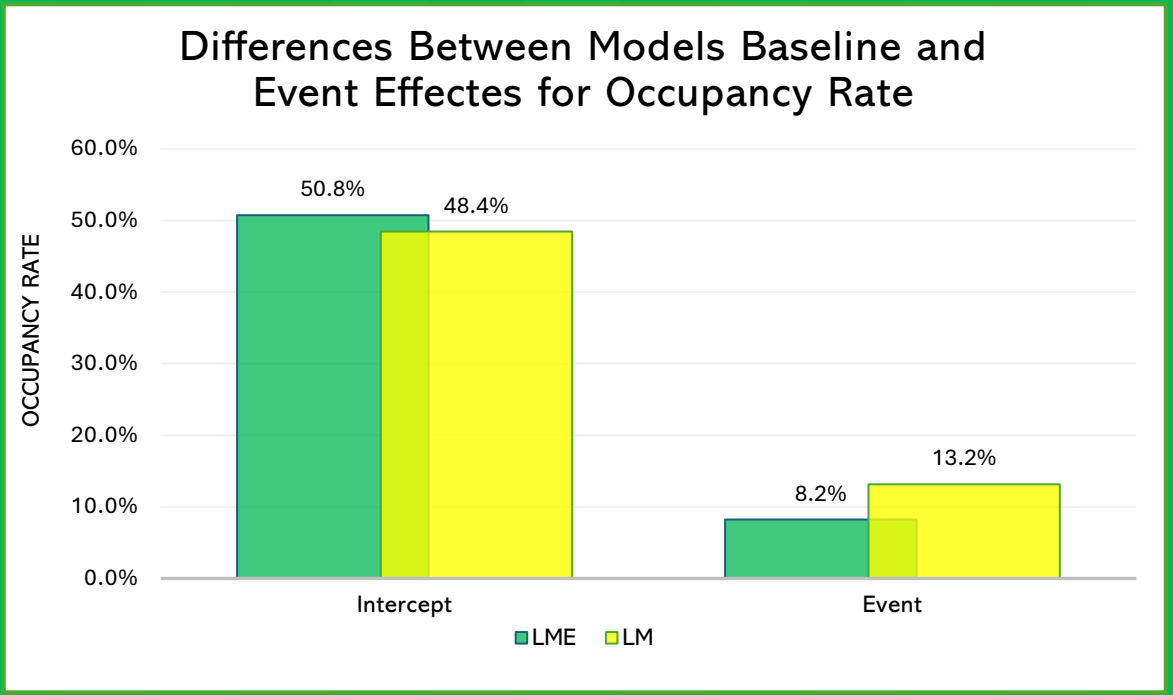
# Occupancy Rate by Cities

- Used LM model with traditional baseline
- LM found the baseline to be 48.4%
- Event Effect to be 13.2%
- Most cities did not statistically differ from baseline (Arlington)
  - Pasadena 3.53 Points
  - New Orleans 1.09 Points
  - Miami -1.95 Points
- Those that did were much lower than the baseline
  - Phoenix -17.63 Points
  - Atlanta - 16.39 Points

```
Call:
lm(formula = OccupancyRate ~ Event * Location + factor(Month) +
    factor(Year) + factor(DayOfWeek), data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-50.093  -6.845   0.157   6.930  54.241

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                48.4476     0.5054  95.862  < 2e-16 ***
Event                      13.1623     4.2460   3.100  0.00194 **
LocationAtlanta             0.4183     0.3341   1.252  0.21070
LocationMiami               7.5425     0.3341  22.572  < 2e-16 ***
LocationNewOrleans         -4.6825     0.3342 -14.012  < 2e-16 ***
LocationPasadena            6.5735     0.3341  19.672  < 2e-16 ***
LocationPhoenix             3.9046     0.3341  11.685  < 2e-16 ***
Event:LocationAtlanta     -16.3906     6.2391  -2.627  0.00862 **
Event:LocationMiami        -1.9491     6.2399  -0.312  0.75477
Event:LocationNewOrleans    1.0883     5.9993   0.181  0.85605
Event:LocationPasadena      3.5307     6.2454   0.565  0.57186
Event:LocationPhoenix     -17.6270     6.2439  -2.823  0.00476 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11.19 on 13471 degrees of freedom
Multiple R-squared:  0.4724,    Adjusted R-squared:  0.4711
F-statistic: 354.8 on 34 and 13471 DF,  p-value: < 2.2e-16
```

# Why do the Models differ

- LMER looks at the effect across cities and is often conservative because of shrinkage
- LM chooses a baseline (Arlington), and the effect is +/-- the baseline
- Even though they are not statistically significant, if you look at the averages from the effects of the LM, the average is close to the 8.24%

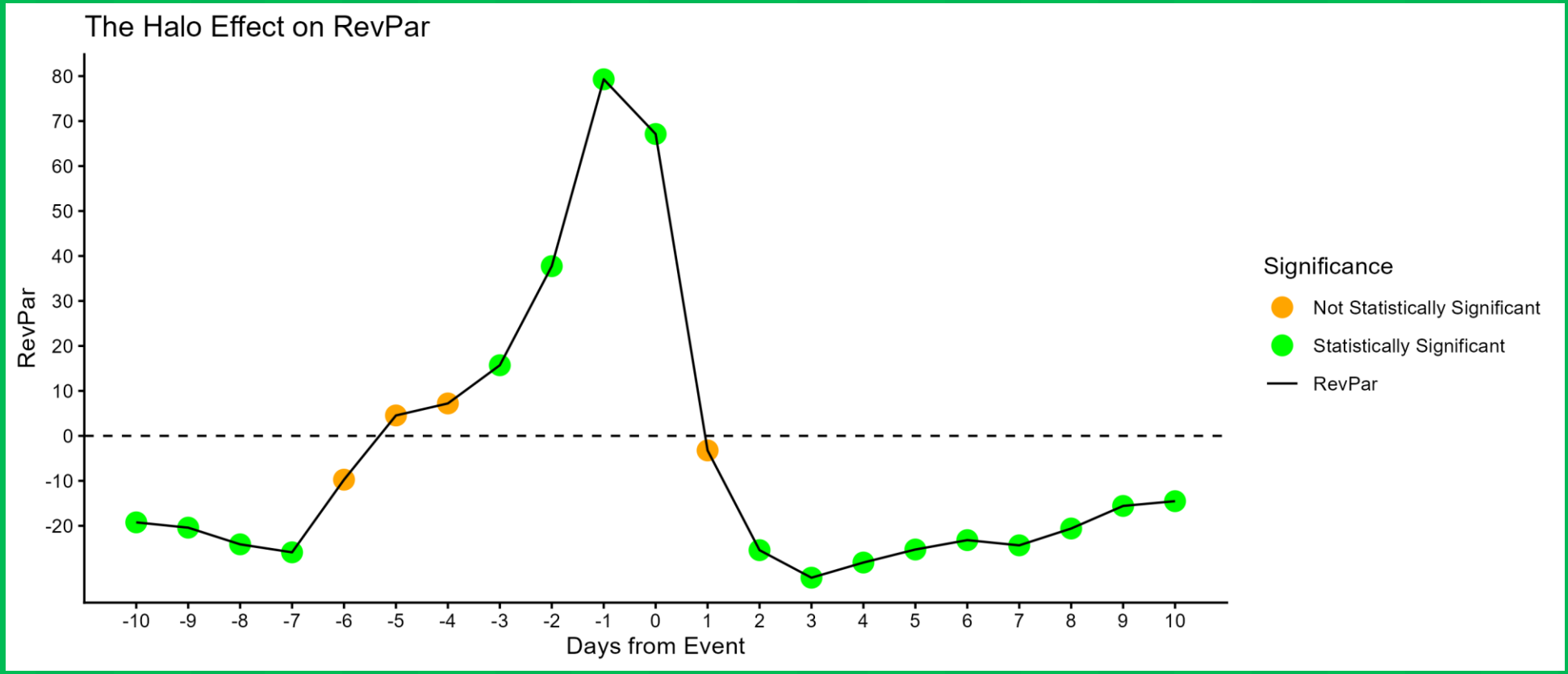



| Baseline (Arlington) | Effect Difference from Baseline | Event Effect* |
|---|---|---|
| 13.16 | N/a | 13.16 |
| 13.16 | -16.39 | -3.23 |
| 13.16 | 1.94 | 15.10 |
| 13.16 | 1.08 | 14.25 |
| 13.16 | 3.55 | 16.72 |
| 13.16 | -17.63 | -4.46 |
| | | |
| | Total Average | 8.59 |

# Halo

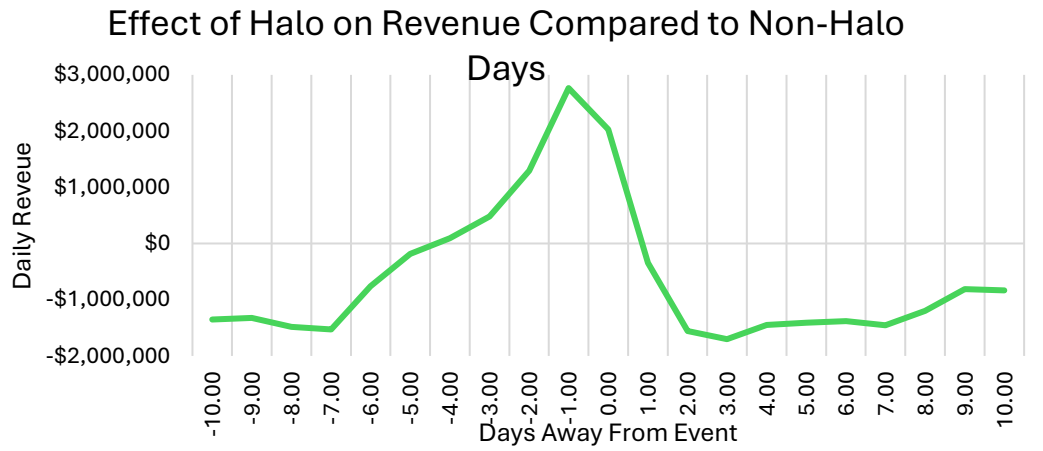
- Viewers of these events are often not just saying for the night of the event, but they say for multiple days.
- The halo is described as the 10 days before the event (-10) and the 10 days after (10)
- We used a linear mixed-effects model that looked at
  - The Halo, Day of Week, Year, Month, and Location were our random effects

# Halo's Effect on RevPar



# Findings

- Very similar to the previous halo graph shape
- Non-halo days may be better because of event spacing. Though the W shape may hold

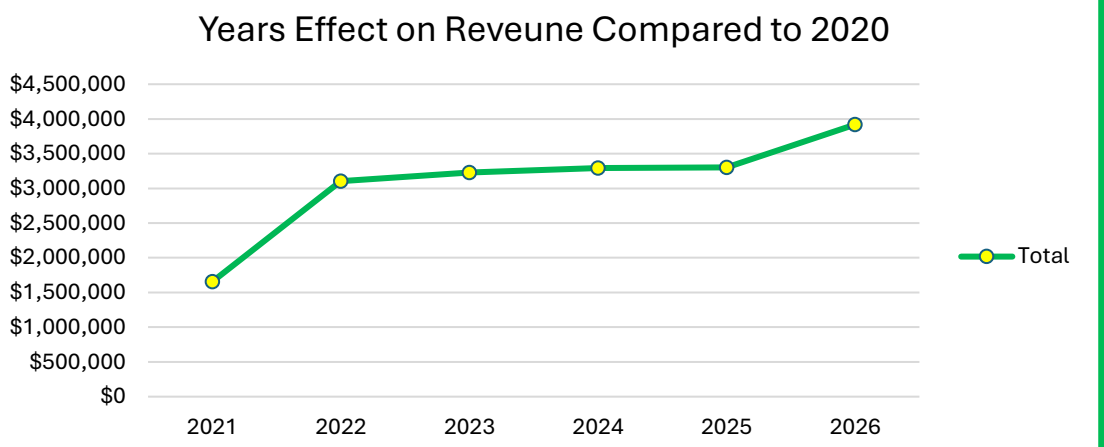



## Halo Revenue

# Findings

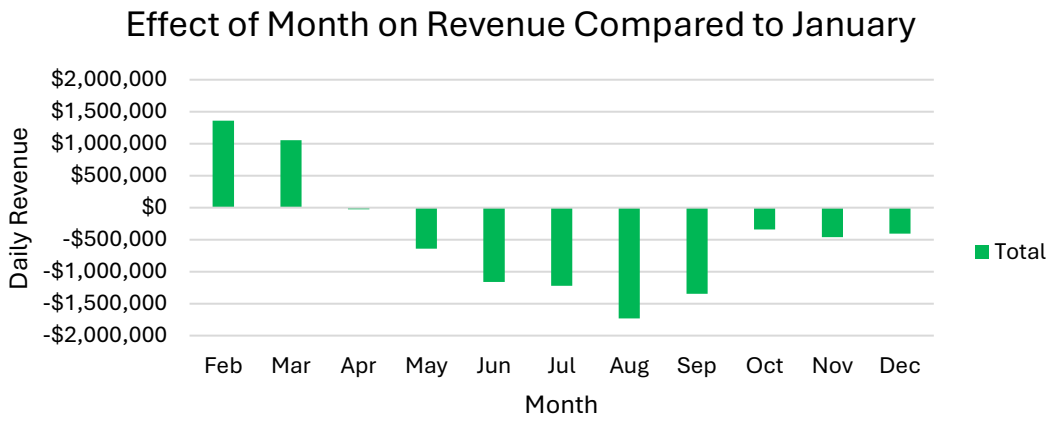
## Years Effect

- Post covid recovery
- With each year having more revenue projected.



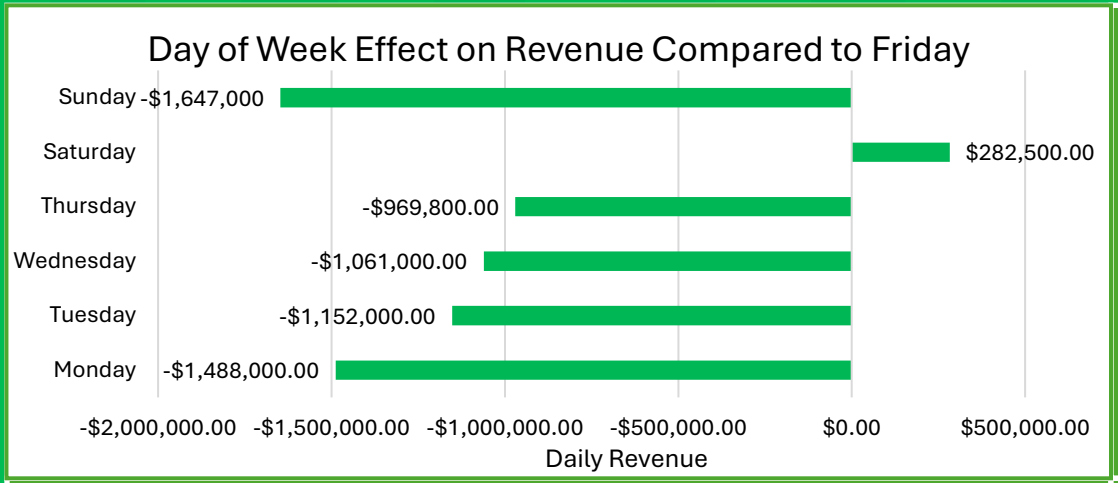
# Findings

## Effect of Month




- Winter months peak revenue as locations are warm southern cities.
- Feb and March are the best, while Aug and Sep are the worst

# Findings




- Negative doesn't mean loss of money, just worse than baseline
- Friday & Saturday are the most profitable days as most visitors are only for a weekend.

## Day of Week Effect

# Economic Effect of Bowl Games on Hotels

- For a **Friday** in **January** in **2025**, that is an **Event Day**

| | |
|---|---|
| • With a baseline daily revenue of | $ 3,770,000 |
| • Event day revenue of | $ 2,031,000 |
| • Halo Revenue of +/- 10 days | $ -706,933 |
| • 2025 Revenue of | $ 3,301,000 |
| • Total Revenue Impact for Hotels | **$ 8,125,007** |

# Recommendations – Hotels

- Hotels in different cities have different needs

- Strong cites: Miami, Pasadena, New Orleans, and Arlington
  - Aggressive pricing because of large increases in occupancy

- Weak Cities: Atlanta and Phoenix: Events have much less effect on Demand, Occupancy Rate, and RevPar
  - Fewer rate hikes, targeted promotions, and focus on competitive pricing

# Recommendations – Hotels Part II

- Plan beyond halo peaks: Multi-night packages throughout halo trough days, along with focusing on Month and Day of Week patterns

- Strong cites: Events cause an increase in occupancy, so increased staffing will be needed

# Recommendations – Organizers

- Treat bowl games as multi-day tourism events, not just a single day
- Create pre- and post-bowl game events like fan events and concerts
- Build new programming around weaker halo days

# Recommendations – Policy Makers

- Cities with Large Bowl Games
  - These bowl games are great for the local economy, and just like event organizers should promote additional tourism around bowl day.
  - Large bowl games will always sell out and make it hard to measure success. Hotel data can help fill the gap in understanding how long the economic prosperity lasts
- Those on the outside looking in
  - Create a friendly location for bowl games to inhabit your city.
  - Good public transit
  - Large Stadium
  - Interesting city culture

# Booking & Bowl Games

- Bowl Games and 20 days around increase hotel revenue by **$8,125,007**
- Events and their Halo effect statistically affect Revenue, RevPar, Demand, and occupancy rate, but not supply.
- Average baseline of **$68.02** RevPar and event increase of **$70.54** RevPar.
- Large Seasonality and baseline variance. Year, Week, Day of Week, and City have large impacts on predictions
- Events do not impact Supply.

Thank you!